

Quiz Submissions - CS571 Summer 2021: Web Technologies Exam 2^x

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Attempt 1

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Submission View

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Cookies and Privacy Questions

Question 1

1.2 / 2 points

[2 points] Cookies include a number of components, some required, some optional. Select only the required components:

- ✓ ☒ Domain
- ✓ ☐ Secure flag
- ✗ ☒ HttpOnly flag
- ✓ ☒ Name / value pair
- ✗ ☐ Expiration date

Question 2

2 / 2 points

[2 points] Select all true properties of cookies. Select all that apply.

Cookies can track your IP address



Http Only cookies are invisible to JavaScript



Persistent cookies may not be deleted



Cookies can store any information you provide to a site



Secure cookies are erased when the user closes the browser

Question 3

1.2 / 2 points

[2 points] Select all ways to "opt out" of cookies. Select all that apply.



Install browser extensions

Technique used to bypass the same-origin policy



Use cookie management tool in browser



Download opt-out cookie



Select "do not track" in browser



Change account Preferences on registration sites

Web Security Questions

Question 4

1.6 / 2 points

[2 points] What are the most recommended ways to generate strong passwords? Select all that apply.



Use a very large number of characters



Use diceware generated passwords

- ✓ ☐ Use passwords that satisfy Microsoft Active Directory settings
- ✓ ☐ Use passwords with low entropy
- ✗ ☒ Use a combinations of 8 or more letters, digits and special characters

Question 5**1.6 / 2 points**

[2 points] Which of the following are Injection Attacks?

- ➡ ✗ ☐ Cross Site Request Forgery (CSRF)
- ➡ ✓ ☒ Search Worms
- ✓ ☐ Content Spoofing
- ➡ ✓ ☒ SQL Injection
- ✓ ☐ DDoS attack

Question 6**1.6 / 2 points**

[2 points] Which types of hacks can cause damage on the client side? Select all that apply.

- ➡ ✓ ☒ Install malicious software
- ✓ ☐ Denial of Service
- ✓ ☐ Deface pages, altering content
- ➡ ✗ ☐ Stealing cookies
- ➡ ✓ ☒ Log user keystrokes

REST Questions

Question 7**1.6 / 2 points**

[2 points] For REST Web Services, select the statements that are true. Select all that apply.

-  ☒ REST services are language dependent
-  ☒ REST services do not offer security
-  ☒ REST services are platform independent
-  ☒ REST services can use HTTPS
- ☒ REST services are a W3C recommendation

High-Performance Websites Questions**Question 8****1.6 / 2 points**

[2 points] Which of the following are ways to reduce the number of HTTP requests? Select all that apply.

-  ☒ Combine Style Sheets
- ☒ Combine cookies
-   ☐ Combine scripts
- ☒ Split images
-  ☒ Use Image Maps

Question 9**1.2 / 2 points**

[2 points] Which of the following are rules to improve Web site performance? Select all that apply.

-  ☒ Reduce number of DOM elements

- ➡ ✖ ☐ Use newer versions of image standards
- ✓ ☐ Use POST for Ajax requests
- ✖ ✓ ☐ Run Google PageSpeed Insights
- ➡ ✓ ✓ ☐ Use cookie-free domains

HTML5 Questions

Question 10

2 / 2 points

[2 points] Which of the following are removed elements in HTML5? Select all that apply.

- ✓ ✓ ☐ center
- ✓ ☐ footer
- ✓ ✓ ☐ font
- ✓ ✓ ☐ frameset
- ✓ ☐ embed

Question 11

1.6 / 2 points

[2 points] Which of these are the 2 most popular HTML5 video codecs? Select only 2.

- ➡ ✓ ✓ ☐ VP9
- ✖ ✓ ☐ Theora
- ✓ ☐ Ogg
- ➡ ✓ ✓ ☐ H.264

✓ ☐ Audio Video Interleave (AVI)

Question 12

2 / 2 points

[2 points] Which of the following attributes are not supported in HTML5? Select all that apply.

✓ ☐ colspan

✓ ☒ bgcolor

✓ ☒ align

✓ ☐ controls

✓ ☒ border

Ajax Questions

Question 13

2 / 2 points

[2 points] The fetch() API can receive cross-site cookies.

☐ True

✓ ☒ False

Question 14

0 / 2 points

[2 points] Creating an instance of the XMLHttpRequest object is the same in all latest browser.

➡ ☐ True

✗ ☒ False

Question 15**2 / 2 points**

[2 points] CORS and server proxy are good ways to avoid cross-domain browser restrictions.

- ✓ ☒ True
☐ False

Question 16**2 / 2 points**

[2 points] The web browser displays a visual progress indicator while the XMLHttpRequest is being processed.

- ☐ True
✓ ☒ False

Question 17**2 / 2 points**

[2 points] Consider the following code snippet:

```
j = eval("(" + text + ")");
```

Select all answers that are TRUE for such code.

- ✓ ☒ It is part of JSON.parse()
✓ ☒ It is very unsafe

Question 18**2 / 2 points**

[2 points] Select all valid XMLHttpRequest() Object properties. Select all that apply.

- ✓ ☐ onReadyState
✓ ☐ readyStateChange
✓ ☒ status

Question 19**2 / 2 points**

[2 points] Of the URLs below, select only the ones that have the same origin as <http://www.ajaxbook.com>?

- ✓ ☐ <http://www.ajaxbook.com:443>
- ✓ ☐ <https://www.ajaxbook.com>
- ✓ ☐ <http://carsearch.ajaxbook.com>
- ✓ ☒ <http://www.ajaxbook.com/ajax>

Question 20**2 / 2 points**

[2 points] Consider the following code from one of the Ajax example shown in class:

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN">
<html><head><title>My Third Ajax Script</title>
<script src="script03.js" type="text/javascript" language="Javascript"></script></head>
<body><div id="pictureBar"> </div></body></html>
```

And this JavaScript:

```
window.onload = initAll;
var xhr = false;
function initAll() { same as previously except it calls getPix }
function getPix() { xhr.open("GET", "flickrfeed.xml", true);
xhr.onreadystatechange = xxxxxx1; xhr.send(null);setTimeout("getPix()",5 * 1000); }
function showPictures() {
    var tempDiv = document.createElement("div");
    var tempDiv2 = document.createElement("div");
    if (xhr.readyState == 4) {
        if (xhr.status == 200) {
            tempDiv.xxxxxx2 = xhr.responseText;
            var allLinks = tempDiv.getElementsByTagName("a");
            for (var i=1; i<allLinks.length; i+=2) {
                tempDiv2.appendChild(allLinks[i].cloneNode(true)); }
            allLinks = tempDiv2.getElementsByTagName("a");
            var randomImg = Math.floor(Math.random() * allLinks.length);
```



```
document.getElementById("pictureBar").innerHTML = allLinks[randomImg].innerHTML;  
    } else { alert("There was a problem with the request " + xhr.status); } } }
```

What should be the values of the two items listed as XXXXX1 and XXXXX2?

✓ ☒ showPictures and innerHTML

☐ callback and loadHTML

Question 21

1.5 / 2 points

[2 points] What are some of the characteristics of Ajax applications? Select all that apply.

➡ ✓ ☒ May use JavaScript to create IFRAMES

➡ ✓ ☒ Interface is manipulated by client-side JavaScript

✗ ☒ Server can manipulate UI using templates

✓ ☐ User interaction is via form submission

Question 22

2 / 2 points

[2 points] Which JavaScript method is used primarily with JSON?

☐ run()

☐ exec()

✓ ☒ eval()

JQuery Questions

Question 23

2 / 2 points

[2 points] Consider the following CSS, jQuery and HTML. Will "Text 2" be changed by the jQuery code, underlined and with green dotted border?

```
<!doctype html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <title>attributeContainsPrefix demo</title>
  <style>
    a {
      display: inline-block;
    }
  </style>
  <script src="https://code.jquery.com/jquery-3.5.0.js"></script>
</head>
<body>

<a href="example.html" hreflang="en">Text 1</a>
<a href="example.html" hreflang="en-UK">Text 2</a>
<a href="example.html" hreflang="english">Text 3</a>

<script>
$( "a[hreflang]='en']" ).css( "border", "3px dotted green" );
</script>
</body>
</html>
```

✓ ☒ True

☐ False

Question 24

2 / 2 points

[2 points] Which of the following selectors are part of the Basic Filter selectors category? Select all that apply.

✓ ☐ :multiple selector

✓ ☒ :header selector

✓ ☒ :first

Question 25

2 / 2 points

[2 points] Which of the following objects are abstracted by jQuery? Select all that apply.

✓ ☒ XMLHttpRequest

✓ ☒ DOM

✓ ☒ JSON

Question 26

2 / 2 points

[2 points] Consider the following example of use of the ":last-child" Selector:

```
$("#input[value='Hot Fuzz'] ").text( "Hot Fuzz" ) ;  
<!doctype html>  
<html lang="en">  
<head>  
  <meta charset="utf-8">  
  <title>last-child demo</title>  
  <style>  
    span.solast {  
      text-decoration: line-through;  
    }  
  </style>  
  <script src="https://code.jquery.com/jquery-3.5.0.js"></script>  
</head>  
<body>
```

```
<div>
  <span>John,</span>
  <span>Karl,</span>
  <span>Brandon,</span>
  <span>Sam</span>
</div>
<div>
  <span>Glen,</span>
  <span>Tane,</span>
  <span>Ralph,</span>
  <span>David</span>
</div>
<script>
$( "div span:last-child" )
.css({ color:"red", fontSize:"80%" })
.hover(function() {
  $( this ).addClass( "solast" );
}, function() {
  $( this ).removeClass( "solast" );
});
</script>
</body>
</html>
```

Which names will be colored red?

 ☒ Sam and David

 ☐ John and Glen

Question 27

2 / 2 points

[2 points] What is the JavaScript object added by jQuery? Select all that apply.

 ☒ \$

 ☒ jQuery

Question 28

2 / 2 points

[2 points] Which of the following selectors are part of the Basic selectors category? Select all that apply.

- ✓ ☒ Element
- ✓ ☐ Form
- ✓ ☒ Multiple

Question 29**2 / 2 points**

[2 points] What is the jQuery code that corresponds to the following DOM code?

```
var kids = n.childNodes;
```

- ☐ kids = \$("body").childNodes();
- ✓ ☒ kids = \$("body").children();

Responsive Website Design (RWD) Questions**Question 30****1 / 2 points**

[2 points] Consider the following media queries:

```
<link rel="stylesheet" media="only screen and (min-width:320px) and (max-width: 760px) "
href="css1.css">
```

```
<link rel="stylesheet" media="only screen and (min-width:761px) and (max-width: 1100px) "
href="css2.css">
```

What different device sizes they fail to support? Select all that apply.

- ➡ ✗ ☐ Devices with large screens

→ ✓ ☒ Devices with small screens

Question 31

2 / 2 points

[4 points] Select which concepts are the basis of Responsive Website Design. Select all that apply.

- ✓ ☐ content reduction
- ✓ ☒ flexible images
- ✓ ☒ fluid grids
- ✓ ☒ CSS3 media queries

Question 32

2 / 2 points

[2 points] Select valid reasons why hosting a .mobi website is not recommended. Select all that apply.

- ✓ ☒ Requires duplication of content
- ✓ ☐ It hinders search engines

Question 33

0.5 / 2 points

[2 points] What are properties of the Responsive Web Design (RWD) approach? Select all that apply.

- ✗ ☒ Uses CSS3 media queries, adaptive grids and flexible images
- ✓ ☒ Adapts to viewing environment
- ✗ ☐ No panning if possible, unless required by the application
- ✗ ☒ Easy reading with a minimum of scrolling

Question 34

2 / 2 points

[2 points] Consider the following code:

```
h1 { margin-left: 14.575%; width: 70.85%; }
```

Is this an example of what kind of dimensions?

- ☐ Pixel-based
- ☐ Rule-based
- ✓ ☒ Relative-based

Frameworks Questions

Question 35

2 / 2 points

[2 points] Which of the following are TRUE of TypeScript? Select all that apply.

- ✓ ☒ Includes support for classes and modules
- ✓ ☐ Supports MVC
- ✓ ☒ Supports Optionals

Question 36

1 / 2 points

[2 points] Select which of the following is true of React. Select all that apply.

- ➡ ✓ ☒ Does not implement MVC
- ✗ ☒ Is a framework like Angular

Question 37**0.667 / 2 points**

[2 points] Select which of the following are TRUE of Angular (version 2 though 9). Select all that apply.

-  ☒ Uses services to increase usability and flexibility
-  ☒ Architecture based on services
-  ☒ Can be used to build Single Page Applications, by rendering plain HTML at the server side

Question 38**1.5 / 2 points**

[2 points] Which of the following are TRUE of Node.js? Select all that apply.

- ☒ Includes built-in CORS support
-  ☒ Core modules include networking and a fetch() compatible API
-  ☒ Includes both synchronous and asynchronous file I/O support
-  ☒ Maintains several connections per server to make HTTP requests

Question 39**1 / 2 points**

[2 points] Select which of the following are TRUE of TypeScript. Select all that apply.

-  ☒ A Semantic superset of JavaScript
-  ☒ Can seamlessly consume JavaScript

Serverless Questions**Question 40****2 / 2 points**

[2 points] Select which of the following is true of AWS Lambda? Select all that apply.

- ✓ ☒ Adapts to rates of incoming requests
- ✓ ☒ Functions are unit of deployment
- ✓ ☐ Can be triggered by HTTP only

Question 41

2 / 2 points

[2 points] Select which of the following are TRUE of AWS Lambda? Select all that apply.

- ✓ ☐ Complex resource model
- ✓ ☒ Easy to build scalable back-end services
- ✓ ☐ Not suitable for real-time data processing

Question 42

2 / 2 points

[2 points] Which of the following are properties of a container like Docker? Select all that apply.

- ✓ ☐ Not portable
- ✓ ☒ Container images can be shared
- ✓ ☒ Containers on same host share single Linux kernel

Question 43

2 / 2 points

[2 points] Which of the following are TRUE of AWS Lambda? Select all that apply.

- ✓ ☒ Functions can be triggered by events
- ✓ ☐ Cost of idle time is small

Functions and classes are unit of deployment



1st Half Course Questions

Question 44

2 / 2 points

[2 Points] Which of the following are valid values of the Accept-Encoding header? Select all that apply.

✓ ☒ compress

✓ ☒ br

Question 45

2 / 2 points

[2 Points] Select all true properties of JSON Objects. Select all that apply.

✓ ☐ Keys are strings or numeric

✓ ☒ Values are JSON values

✓ ☒ Objects are unordered containers of key/value pairs

✓ ☒ Can be used to represent: struct, record, hashtable, object

Question 46

1.333 / 2 points

[2 Points] Below are lines extracted from the CSS style sheet of google.com.

```
.gb_N .gb_M {  
    background-image: none !important;  
    position: relative  
}
```

Which of the following examples will trigger the CSS?

- ✓ ☒ `<div class="gb_N">`
 `<div class="gb_M">`
 `. . .`
 `<div>`
 `</div>`
- ✓ ☐ `<div id="gb_N">`
 `<div id="gb_M">`
 `. . .`
 `<div>`
 `</div>`
- ✗ ☒ `<body class="gb_N">`
 `. . .`
 `<div>`

Question 47

2 / 2 points

[2 Points] Select characters that to appear in HTML need to be represented with numeric or entity references. Select all that apply.

- ✓ ☒ & (ampersand)
- ✓ ☒ " (double quote)
- ✓ ☒ ' (single quote)

Question 48

2 / 2 points

[2 Points] Consider the following program:

```
alert( "123".match( /\d/g ))
```

What is displayed by the alert?

☐ 1

✓ ☒ 1,2,3

Question 49

2 / 2 points

[2 Points] Consider the following Python code:

```
>>> a = set('abracadabra')  
>>> b = set('alacazam')  
>>> a & b
```

What gets printed?

☐ {'a', 'b', 'c', 'd', 'l', 'r', 'm', 'z'}

☐ {'a', 'b'}

✓ ☒ {'c', 'a'}

Question 50

2 / 2 points

[2 Points] Consider the following example from w3schools.com.

```
<!DOCTYPE html>  
<html>  
<body>  
<p>Click the button.</p>  
<button onclick="myFunction()">Try it</button>
```

```
<script>
function myFunction() {
var btn = document.createElement("BUTTON");
btn.innerHTML = "CLICK ME";
document.body.appendChild(btn);
}
</script>
</body>
</html>
```

What happens when one clicks the "Try it" button?

- ☐ Creates a BUTTON element with text "CLICK ME" above the text "Click the button".
- ✓ ☒ Creates a BUTTON element with text "CLICK ME" next to the "Try It" button
- ☐ Creates a BUTTON element with text "CLICK ME" above the "Try It" button
- ☐ Creates a BUTTON element with text "CLICK ME" below (i.e., under) the "Try It" button

Attempt Score:85.7 / 100 - 85.7 %

Overall Grade (highest attempt):85.7 / 100 - 85.7 %

Done